

Match Prediction and Relationship Preference Analysis

Using the Columbia University Speed Dating Dataset

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Abstract—This report investigates mutual romantic match prediction in speed dating as a supervised binary classification task, using 8,378 interaction records from the Columbia University Speed Dating dataset. Because the same participant appears multiple times in different interaction records, traditional row-level random splitting will cause severe participants' data leakage and over-optimistic evaluation results. To address this, we propose a leakage-safe wave-level participant-disjoint splitting strategy, selecting the optimal partition via 5,000-iteration Monte Carlo search to optimise test-set size and class distribution while guaranteeing zero participant overlap after splitting.

Drawing on behavioural findings from speed dating literature, we develop a relationship-aware feature engineering framework transforming 195 raw attributes into 54 interaction-level features covering five core relational mechanisms. We adopt the Global Persona Baseline (GPB) normalisation to eliminate individual scoring-scale bias, and find cosine similarity on GPB-normalised interest vectors achieves the strongest correlation with match outcomes ($|r| = 0.1071$), outperforming alternative metrics and the raw baseline.

Three models are implemented entirely from scratch in NumPy: Logistic Regression, AdaBoost, and RBF-kernel SVM. Logistic Regression achieves the best overall performance (test $F_1 = 0.5632$, AUC = **0.8474**), followed closely by AdaBoost ($F_1 = 0.5538$, AUC = **0.8410**) and SVM ($F_1 = 0.5367$, AUC = **0.8229**). Post-hoc analysis confirmed that in this unbalanced behavior prediction task, the performance of the participants' original personal information was not as good as that of the artificially constructed features that focused on "interactivity".

Index Terms—speed dating, match prediction, feature engineering, class imbalance, logistic regression, AdaBoost, support vector machine

I. INTRODUCTION

A. Problem Statement

Predicting whether two people will mutually choose to pursue a romantic connection is a problem at the intersection of behavioural science and applied machine learning. In speed dating, each participant has brief, fixed-format interactions with a series of partners. During this process, each participant independently records their impression of each date and their own decision on whether to pursue the relationship. In speed dating, a mutual match occurs only when both participants simultaneously express a wish to meet again. This bilateral structure makes the task of predicting matching outcomes fundamentally different from standard unilateral preference modeling. In unilateral preference modeling, we only look at the ratings one person gives to another. However, here, if only

one party gives a very high evaluation, its predictive value is limited, because if the other party's evaluation is not high, then the two people will still not match successfully in the end. Formally, this problem is a binary classification task in which a model will receive a feature vector that describes the information of a pair of participants and the interaction between them as input. The model must then predict a binary matching result using only information available at the time of the date, and cannot use information learned after the fact.

B. Motivation and Background

Speed dating experiments offer a rare and controlled setting in which real-time partner preferences can be observed and measured. In contrast, the vast majority of studies on the choice of romantic partners rely on the results of questionnaire survey data or online personal profiles. When people fill out questionnaires or design personal profiles, they may subconsciously beautify themselves by saying or showing answers that "society deems right" or "make them seem good", rather than their true inner thoughts. They might also deliberately present themselves inauthentically for some purpose. The Face-to-face speed dating experiment, taking place in real and immediate interactions, allows participants to avoid excessive embellishment, thus enabling the capture of more genuine and natural choice preferences. From the perspective of practical application, understanding which features can predict successful matches can help design more effective and accurate recommendation systems for move beyond simplistic matching on stated preferences.

The Columbia University Speed Dating dataset, introduced by Fisman et al. [1], is the canonical resource for this research area. It was collected over several consecutive years through structured speed dating activities at the Columbia Business School. During the event, participants will be randomly paired and take turns after a brief four-minute exchange.

After each encounter, both participants completed a detailed questionnaire rating their partner on five core attributes (attractiveness, sincerity, intelligence, humour, and ambition), indicating overall liking, estimating the probability that the partner liked them, and reporting their match decision. The dataset also includes extensive demographic information, self-assessed trait ratings, and ratings of 17 leisure interest cate-

gories, making it well-suited for studying the interaction among various predictive signals

C. Challenges and Contributions

There are three substantive challenges must be addressed. **To begin with**, the class imbalance is severe with only approximately 16.5% of encounters result in a mutual match (as shown in Fig. 1). A classifier that always predicts no match achieves 83.53% accuracy without providing any useful information. Standard accuracy-based model selection is therefore misleading, and F_1 -score and AUC-ROC must be adopted in the evaluation part.

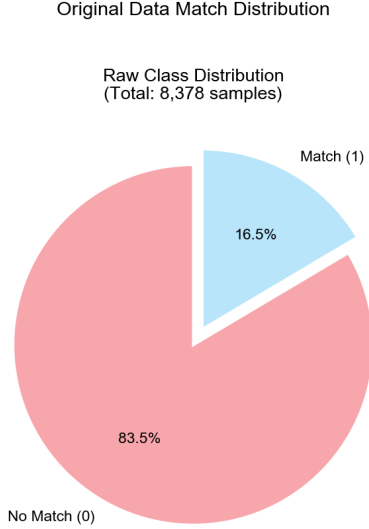


Fig. 1. Class distribution: 16.5% mutual matches vs. 83.5% non-matches.

Furthermore, the dataset has a hierarchical structure in which each participant appears across many rows. If the data is simply randomly split by rows when being divided, the multiple appointment records of the same participant will be distributed to both the training set and the test set. This will lead to data leakage, and the final performance estimate obtained will be overestimated and not objective enough. This data leakage must be addressed through a participant-disjoint splitting strategy.

Finally, raw features are inherently individual-level. Since the match is jointly determined by both sides, it depends on whether their mutual evaluations are consistent. Therefore, it is necessary to artificially construct the features of the relationship hierarchy to clearly represent the similarities between the two sides, their mutual attraction and equivalent interests, so that the model can utilize all the predictive signal.

This report makes four primary contributions: (1) a wave-level, participant-disjoint splitting strategy that prevents leakage, with the optimal wave assignment identified via a 5,000-iteration Monte Carlo search; (2) a domain-grounded feature engineering pipeline producing 54 relationship-level features validated post-hoc by correlation analysis, including a GPB-normalised cosine interest-similarity measure that outperforms

both the SIS baseline and the raw feature; (3) from-scratch NumPy implementations of three classification models; and (4) a systematic empirical comparison covering performance metrics, threshold sensitivity, feature importance, and computational cost.

II. RELATED WORK AND EXISTING TECHNIQUES

Fisman et al. [1] introduced the Columbia Speed Dating dataset and explored gender differences in mate selection preferences. Their research found that men attach significantly more importance to physical attractiveness than women, while women place relatively greater emphasis on intelligence and shared interests. This conclusion prompted us to incorporate gender-by-trait interaction features into our modelling pipeline.

Luo and Zhang [2] confirmed that the strongest single indicator in predicting whether a person will develop a romantic attraction to another is whether the person senses that their emotions have been responded to by the other party. They call this perceived reciprocal affection, and its predictive power significantly exceeds that of physical attraction and personality similarity. This discovery directly inspired us to construct the interaction feature of *like* $\times$ *guess_prob_liked*. Our further analysis verified that this feature belongs to the category of the most predictive features (Pearson correlation coefficient $r = 0.3247$).

Zhao et al. [4] pointed out that there exists a systematic deviation between the ideal partner preferences self-reported by respondents in advance and their actual dating behaviour. This is because real-world decision-making integrates contextual information and relational cues that participants cannot anticipate through pre-event questionnaires. This finding guided our decision to prioritise post-date mutual evaluation scores over pre-date stated preference weights.

From the perspective of machine learning, Guo et al. [3] applied standard classifiers such as logistic regression, decision tree, random forest, and gradient boosting to the speed matching prediction task, and the reported AUC values ranged from 0.80 to 0.87 depending on configuration. Our work extends these findings in three directions:

- 1) All models are implemented entirely from scratch rather than calling third-party machine learning libraries;
- 2) We address the participant-level data leakage problem that was not discussed in prior work;
- 3) We design a customised feature engineering pipeline grounded in behavioural science domain theory, rather than applying generic feature selection methods to raw attributes.

III. METHODOLOGY

A. Overview of the Proposed Framework

The overall workflow of the proposed framework is illustrated in Fig. 2. The pipeline begins with leakage-safe wave-level participant-disjoint splitting, followed by relationship-aware feature engineering and train-only feature selection. Three machine learning models—Logistic Regression, AdaBoost, and SVM—are subsequently trained and evaluated

Speed Dating Match Prediction Pipeline

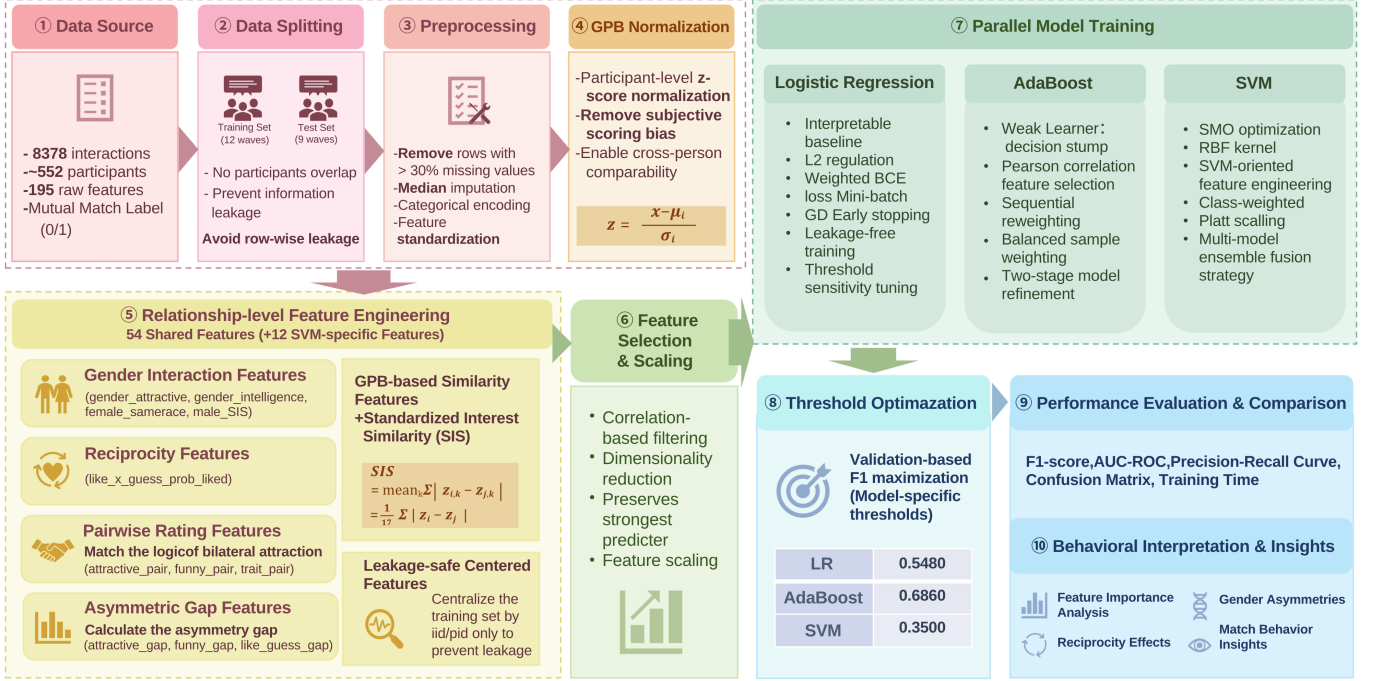


Fig. 2. Overall machine learning pipeline: from leakage-safe wave-level splitting through relationship-aware feature engineering, model training, and threshold-optimised evaluation.

under threshold-optimised settings. This design ensures that all preprocessing, tuning, and evaluation procedures remain free from participant-level information leakage.

B. Dataset and Preprocessing

The Columbia University Speed Dating Dataset contains repeated interpersonal interactions collected during real speed dating events. In total, the dataset comprises 8,378 pair-level records from approximately 552 unique participants, described by 195 original features grouped into seven thematic categories (Fig. 3).



Fig. 3. Distribution of 195 original features across seven thematic categories.

Each participant evaluated multiple partners in the same group as themselves through dimensions such as attractiveness,

intelligence, sincerity, fun, ambition, and shared interests.

One of the main challenges of this dataset lies in the fact that the participants' ratings are essentially subjective and lack consistency across individuals. Some participants tended to give high scores to most of their partners, while others were more conservative in their ratings. The same values might represent completely different underlying preferences. Therefore, the original scores cannot be directly compared among the participants.

To address this issue, before feature engineering and model training, we introduced a participant-centred normalization framework called **Global Personality Baseline (GPB)**. GPB standardizes the ratings within each participant, thereby eliminating subjective rating preferences while retaining their relative preference structures:

$$z_i = \frac{x_i - \mu_p}{\sigma_p} \quad (1)$$

where x_i is the original rating, μ_p the participant-specific mean, and σ_p the participant-specific standard deviation. This transformation transforms the original assessment into a relative preference intensity, thereby making comparisons across participants more meaningful.

After GPB normalisation, interest similarity is measured using three complementary metrics applied to the GPB-normalised interest vectors $\mathbf{z}^{(A)}$ and $\mathbf{z}^{(B)}$ of each pair.

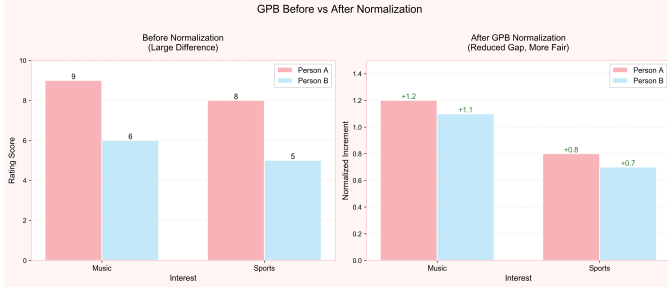


Fig. 4. GPB normalisation: raw rating distributions (left) vs. standardised distributions (right).

The **Standardised Interest Similarity (SIS)** is derived from the mean absolute difference between normalised scores:

$$\text{SIS} = 1 - \frac{1}{n} \sum_{i=1}^n |z_i^{(A)} - z_i^{(B)}| \quad (2)$$

SIS measures the matching degree of absolute enthusiasm between two participants in various areas of interest. Despite its intuitive appeal, it yields $|r| = 0.0122$ with match outcomes—slightly *below* the raw `interests_correlate` baseline ($|r| = 0.0309$). This indicates that merely standardizing the scoring bias cannot restore useful signals from intensity-based comparisons

A **variance-weighted Euclidean similarity** is also computed. This method assigns higher weights to interests with larger variances at the overall level, while assigning lower weights to interest with smaller variances. The basic intuition is that interests with high variance are more distinguishable and thus more informative for measuring compatibility. The measure achieves $|r| = 0.0206$, modestly above SIS but still substantially below the raw baseline.

The **Cosine similarity** of the GPB standardized interest vector captured the directional consistency between the *preference profiles* of two participants, that is, the relative ranking and priority of their interests, regardless of their absolute enthusiasm level:

$$\text{sim}_{\cos} = \frac{\mathbf{z}^{(A)} \cdot \mathbf{z}^{(B)}}{\|\mathbf{z}^{(A)}\| \|\mathbf{z}^{(B)}\|} \quad (3)$$

This distinction is behaviourally meaningful. If two participants prioritize the same interests at their top, then even if one shows general enthusiasm in all aspects while the other is more picky, they still have the same preference *direction*. Cosine similarity precisely captures this structural similarity while keeping the scoring scale for individuals unchanged. Empirically, cosine similarity achieves $|r| = 0.1071$, compared with $|r| = 0.0309$ for the raw `interests_correlate` baseline. This represents an increase of approximately 246% and is significantly better than both SIS and variance-weighted variants.

This result confirms two things: (i) GPB standardization is a necessary prerequisite, as directly calculating cosine similarity on the raw score would confuse preference structure with rating

scale bias; (ii) In interest-based compatibility, the direction consistency of preference profiles, rather than the degree of intensity proximity, is the dominant factor. Fig. 4 shows the standardized distribution effect of GPB, illustrating how the multi-modal, person-anchored raw score distribution converges into a unimodal standardized distribution after centralization within the participants.

The dataset also contains substantial missing data due to incomplete questionnaires and participant nonresponse. Sample rows where the proportion of missing values in the feature vector exceeds 30% are removed, and the remaining numerical missing data are filled with the median of the training set to prevent information leakage. Categorical variables are numerically encoded, and continuous variables are standardized where appropriate. All imputation statistics are calculated only on the training set and then applied to the validation set and test set, thereby ensuring that the information in the test set does not affect the preprocessing flow.

Leakage-safe splitting. Under naïve row-level splitting, the same participant may appear in both training and testing sets through multiple dating interactions, artificially inflating predictive performance. To avoid this situation, each of the dating waves was fully allocated to the training set, validation set or test set to ensure that there is no overlap of participant information among the splits as shown in Fig. 5.

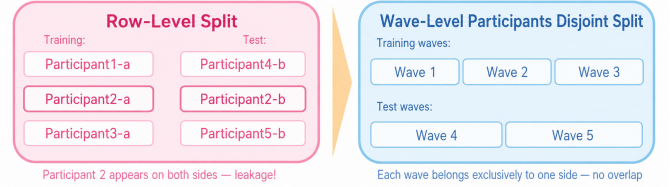


Fig. 5. Wave-level participant-disjoint data splitting vs. naïve row-level splitting.

How to allocate each round to different data partitions is not a simple combinatorial optimization problem, because waves vary in size and match rate. If a simple and crude distribution method is adopted, there may be significant differences in the category distribution between the test set and the training set, which in turn affects the reliability of the evaluation results. To address this issue, we conducted a **5,000-iteration Monte Carlo random search** on all feasible round allocation schemes and optimized a composite function containing two terms: (i) the square deviation between the proportion of the test set and the target value of 30%. (ii) the absolute difference in the positive matching rate between the training set and the test set. The second objective was set with double weights because the F_1 -score and AUC are more sensitive to the distribution changes of the minority classes. Ultimately, we chose the allocation scheme that minimizes this composite objective function as the final data partitioning method. The retained test set obtained according to this scheme contains 2,449 interactions, and the positive matching rate is 16.90%.

The positive matching rate of the training set is 16.89%, with a difference of less than 0.01 percentage points between the two. More importantly, there is *no overlap* of participants at all between the two data partitions. In previous studies that used this dataset, it was rare to achieve such good distribution alignment and such strict participant isolation.

Since successful matches only account for a small portion of all interactions, this dataset still has a highly imbalanced problem. Therefore, the assessment mainly focuses on F_1 -score and AUC, rather than the overall accuracy rate.

C. Feature Engineering

In the entire framework, feature engineering occupies the most important position. The information provided by the original dataset mostly remains at the individual level, that is, the one-sided evaluation of participant A on participant B. However, whether a two-way match can be formed essentially depends on whether both sides appreciate each other and are in tune with each other. This is a bilateral outcome. To more accurately depict this relationship structure, based on the original scores, we designed five types of artificial features that can reflect the interactive relationship.

Gender Interaction Features [1] model asymmetric weighting of mate preferences across genders:

$$\begin{aligned} \text{gender_attractive} &= \text{gender} \times \text{attractive}, \\ \text{gender_intelligence} &= \text{gender} \times \text{intelligence}. \end{aligned}$$

These terms allow the model to capture gender-dependent valuation patterns in attractiveness and personality traits.

Reciprocity Features [2] quantify mutual romantic reinforcement. The most important reciprocity feature is:

$$\text{like_x_guess_prob_liked} = \text{like} \times \text{guess_prob_liked}$$

This feature achieves post-hoc correlation $|r| = 0.3247$, substantially outperforming most raw individual ratings.

Pairwise Rating Product Features model bilateral attraction intensity by multiplying self-to-partner and partner-to-self evaluations:

$$\begin{aligned} \text{attractive_pair} &= \text{attractive} \times \text{attractive_o}, \\ \text{funny_pair} &= \text{funny} \times \text{funny_o}, \dots \end{aligned}$$

Among all engineered variables, *attractive_pair* exhibits the strongest correlation with match outcome ($|r| = 0.3973$).

Asymmetric Evaluation Gap Features measure disagreement between reciprocal evaluations:

$$\text{attractive_gap} = |\text{attractive} - \text{attractive_o}|, \dots$$

Large gap values indicate asymmetric attraction or imbalance in interpersonal perception.

GPB Normalisation + Cosine Similarity address interpersonal scoring bias as described in Section III-B. Raw interest-related attributes are first transformed into within-participant standardised scores using GPB normalisation, after which three pairwise similarity metrics are computed and compared: SIS ($|r| = 0.0122$), variance-weighted Euclidean

similarity ($|r| = 0.0206$), and cosine similarity ($|r| = 0.1071$). Notably, SIS falls *below* the raw *interests_correlate* baseline ($|r| = 0.0309$), while cosine similarity achieves a 246% improvement, confirming that directional preference alignment is the operative compatibility signal.

The complete shared engineered feature space contains 54 features. For the SVM model, 12 additional interaction features expand the final input space to 66 dimensions. These features mainly capture reciprocal attraction and compatibility patterns (Section III-F). Table I summarises all five categories.

TABLE I
SUMMARY OF FEATURE ENGINEERING CATEGORIES

Category	Key Features	Rationale	Source
Gender Inter.	gender_attr, gender_intel	Trait weights differ by gender	[1]
Reciprocity	like × guess_prob	Perceived mutual liking predicts match	[2]
Pairwise Prod.	attr_pair, funny_pair, trait_pair	Mutual high ratings reflect bilateral appeal	[1]
Gap Features	attr_gap, funny_gap, like_guess_gap	One-sided interest predicts failure	—
GPB + SIS	SIS, int_cos, int_euc_sim	Bias-corrected shared interest similarity	This work

D. Logistic Regression

Logistic Regression serves as the interpretable baseline model for the match prediction task. Given a relationship-level feature vector $\mathbf{x}$, the model estimates the probability of a successful match through a sigmoid-transformed linear combination:

$$P(y=1 | \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (4)$$

The model is trained using weighted binary cross-entropy with ℓ_2 regularisation:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N \alpha_i \left[y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i) \right] + \frac{\lambda}{2N} \|\mathbf{w}\|_2^2 \quad (5)$$

Here, α_i denotes the class-dependent sample weight, and the intercept term is excluded from regularisation. To address the strong class imbalance in the Speed Dating dataset, balanced class weights are used, and an additional positive-class weight scaling factor is tuned over $\{0.7, 0.8, 0.9, 1.0\}$ to balance precision and recall.

The model was implemented from scratch using NumPy and trained with mini-batch gradient descent with momentum. Hyperparameters, including the learning rate, regularisation strength, batch size, and positive-class weight scaling factor, were selected using five-fold stratified cross-validation. Three feature configurations were compared: all available engineered features, a re-selected Pearson-correlation Top-18 subset, and

an older fixed Corr18 baseline. The final reported Logistic Regression model uses the re-selected Corr Top18 feature subset.

A key design choice is threshold optimisation. Since successful matches account for only about 16.9% of the processed training data, the default threshold of $\tau = 0.5$ is not necessarily optimal for maximising minority-class performance. Therefore, the classification threshold was tuned within five-fold stratified cross-validation by maximising validation F_1 -score on each fold, and the final threshold was obtained by averaging the fold-wise optimal thresholds. The selected threshold was $\tau = 0.548$, confirming that decision thresholds should be treated as tunable hyperparameters in imbalanced prediction tasks.

An additional advantage of Logistic Regression is interpretability: the learned coefficients indicate the direction and relative strength of association between each feature and match probability, helping identify which engineered interaction features contribute most strongly to romantic compatibility prediction.

E. AdaBoost with Decision Stumps

AdaBoost (Adaptive Boosting) is an ensemble learning method that sequentially combines multiple weak learners into a stronger classifier. In this project, each weak learner is implemented as a decision stump, which is a single-split decision tree based on one feature threshold. Compared with Logistic Regression, AdaBoost with decision stumps can capture nonlinear threshold effects and combine multiple engineered relationship-level signals through a weighted sequential ensemble while maintaining the computational efficiency of tabular data.

Formally, the binary labels are converted into signed labels:

$$y_i \in \{-1, +1\}$$

At boosting round t , a decision stump $h_t(x)$ is trained to minimize the weighted classification error:

$$\epsilon_t = \sum_{i=1}^n w_i^{(t)} \cdot \mathbb{I}[h_t(x_i) \neq y_i]$$

where $w_i^{(t)}$ is the sample weight of observation i at round t , and $\mathbb{I}(\cdot)$ is the indicator function. The contribution weight of the stump is then computed as:

$$\alpha_t = \eta \cdot \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

where η is the learning rate. Weak learners with weighted error greater than or equal to 0.5 are not added to the ensemble, because they perform no better than random guessing.

After each boosting round, sample weights are updated as:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \cdot \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

where Z_t is a normalization factor:

$$Z_t = \sum_{i=1}^n w_i^{(t)} \cdot \exp(-\alpha_t y_i h_t(x_i))$$

This update decreases the weights of correctly classified samples and increases the weights of misclassified samples, forcing later stumps to focus more strongly on difficult cases.

The final ensemble score is obtained by a weighted sum of all stump predictions:

$$F(x) = \sum_{t=1}^T \alpha_t h_t(x)$$

In standard AdaBoost, the final signed prediction can be written as:

$$\hat{y} = \text{sign}(F(x))$$

In our implementation, however, $F(x)$ is transformed into a probability-like score:

$$p(y = 1 | x) = \frac{1}{1 + \exp(-2F(x))}$$

The final class label is assigned using a validation-optimized threshold τ :

$$\hat{y} = \begin{cases} 1, & p(y = 1 | x) \geq \tau \\ 0, & p(y = 1 | x) < \tau \end{cases}$$

In order to solve the problem of class imbalance, our implementation uses class-balanced initial sample weights, so that before boosting begins, positive and negative samples receive equal total weight. Then, AdaBoost further updates the weight of a single sample according to the misclassification. This design combines explicit imbalance handling with AdaBoost's adaptive weight mechanism to help later weak learners focus more on difficult positive match cases without synthetic oversampling.

Before training, use only training based on correlation-based feature selection, and the dimension is reduced from 54 characteristics. Instead of manually fixing the feature subset size, the feature subset size and AdaBoost hyperparameters were selected through repeated stratified 5-fold cross-validation. Specifically, we used 10 different random seeds for the stratified folds, resulting in 50 validation folds for each configuration. This repeated cross-validation design reduces the instability caused by a single random fold split, which is especially important because the F1-score is sensitive to threshold selection under severe class imbalance.

The search space included TopK feature subsets from Top25 to Top35, number of estimators $T \in \{50, 70, 90, 110\}$, and learning rate $\eta \in \{0.1, 0.3, 0.5\}$. The best configuration selected by mean CV-F1 used Top35 correlation-selected features, 110 decision stumps, and learning rate $\eta = 0.50$. Threshold optimization was also performed within validation folds, and the final threshold was set to $\tau = 0.660$. Since this threshold is higher than the default 0.5, the final classification decision uses the validation-optimized threshold rather than a fixed default cutoff.

The selected Top35 feature subset preserves a broader set of relationship-level engineered variables, including pairwise rating products, centered rating features, gap features, interest similarity features, and gender interaction terms. This design

allows the AdaBoost model not only to use the original personal rating, but also to use reciprocity and relationship signals that are more directly consistent with the mutual nature of matching prediction tasks.

F. Support Vector Machine with RBF Kernel and SMO Optimisation

SVM was implemented to model nonlinear interaction patterns in the speed dating dataset. Compared with Logistic Regression, the RBF kernel can better capture complex relationships between attraction, shared interests, and perceived mutual liking. The final SVM input contains 66 features, including 12 additional engineered interaction features.

The Radial Basis Function (RBF) kernel was adopted:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2) \quad (6)$$

To address class imbalance, larger penalty weights were assigned to positive-match samples during training. This improves minority-class sensitivity without additional resampling. Hyperparameters, including C , γ , and positive-class weight multipliers, were selected through validation-based search.

Because raw SVM scores are not calibrated probabilities, Platt scaling was applied on validation data before threshold evaluation. The optimal threshold was $\tau = 0.340$, favouring higher recall under the imbalanced match distribution.

Multiple candidate SVM models were evaluated and combined through validation-based ensemble selection. Although the final SVM ensemble achieved competitive performance, the results suggest that feature engineering contributes more to prediction performance than additional model complexity.

IV. EXPERIMENTAL STUDY AND RESULTS ANALYSIS

A. Evaluation Protocol

All hyperparameter tuning, threshold optimization and feature selection processes are carried out only within the training set, using internal validation splitting or grid search based on cross-validation, while the retained test set is accessed only once during the final evaluation.

To ensure the authenticity of the generalization ability assessment, we adopted a participant non-overlapping splitting strategy. The entire speed matching round is fully allocated to the training set or reserved test set. This round-by-round splitting avoids information leakage at the participant level. This is a key issue that has been overlooked in previous studies. If leakage occurs, it will lead to overly optimistic performance evaluation results of the model.

The retained test set contains 2,449 interactions, with a positive matching rate of 16.90% (16.89% in the training set), which reflects the natural fluctuations between different speed matching rounds. Under such a severe category imbalance, accuracy loses its reference value: a simple classifier that always predicts "nomatch" can achieve an accuracy rate of approximately 83.5%. Therefore, the evaluation mainly focuses on the F_1 -score and AUC-ROC rather than the overall accuracy rate. The F_1 -score balances the precision and recall rates of minority class predictions, while the AUC assesses the

ranking quality that does not rely on classification thresholds. In addition, we also reported the precision-recall (PR) curve, as it can provide more intuitive visual information than a single ROC curve in the case of data imbalance.

This evaluation process ensures that the reported performance reflects the true generalization ability, rather than the artificial results caused by overlapping participants or improper evaluation metrics.

B. Overall Performance Comparison

Table II summarises the test-set performance of all three models.

TABLE II
TEST-SET PERFORMANCE OF ALL THREE MODELS

Model	Acc.	Prec.	Rec.	F_1	AUC	τ	Time
LR	0.802	0.450	0.754	0.563	0.8474	0.548	0.35s
ADA	0.809	0.457	0.703	0.554	0.8410	0.660	3.06s
SVM	0.820	0.474	0.618	0.537	0.8229	0.340	34.07s

LR=Logistic Regression, ADA=AdaBoost, SVM=Top-5 ensemble.

The overall comparative performance is visualised in Fig. 6.

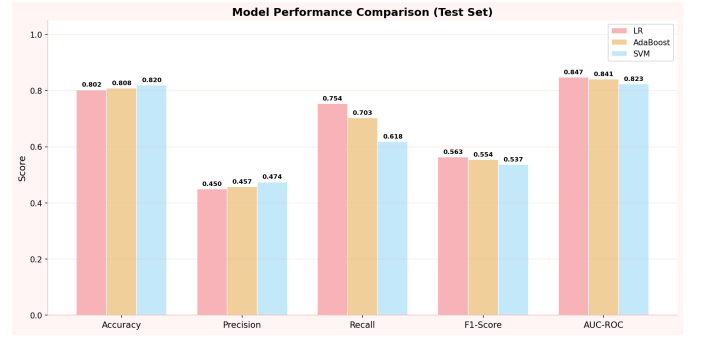


Fig. 6. Performance comparison (Accuracy, Precision, Recall, F_1 , AUC) across all three models.

Logistic Regression achieves the strongest overall result, obtaining an F_1 -score of 0.5632 and an AUC of 0.8474. AdaBoost follows closely ($F_1 = 0.5538$, AUC=0.8410), while the SVM ensemble achieves $F_1 = 0.5367$ and AUC=0.8229.

Receiver Operating Characteristic (ROC) curves are shown in Fig. 7.

Logistic Regression marginally outperforms AdaBoost in global discrimination ability, although both models demonstrate similarly strong discrimination performance across thresholds. The SVM ensemble achieves competitive performance, although its computational cost is much higher because of kernel computation and validation-based model selection. However, because the dataset is heavily imbalanced, ROC analysis alone may overestimate practical classification effectiveness. Precision-Recall curves therefore provide a more informative evaluation of positive-class retrieval behaviour (Fig. 8).

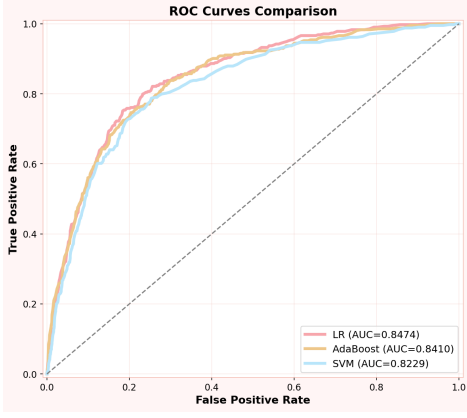


Fig. 7. ROC curves for Logistic Regression, AdaBoost, and SVM.

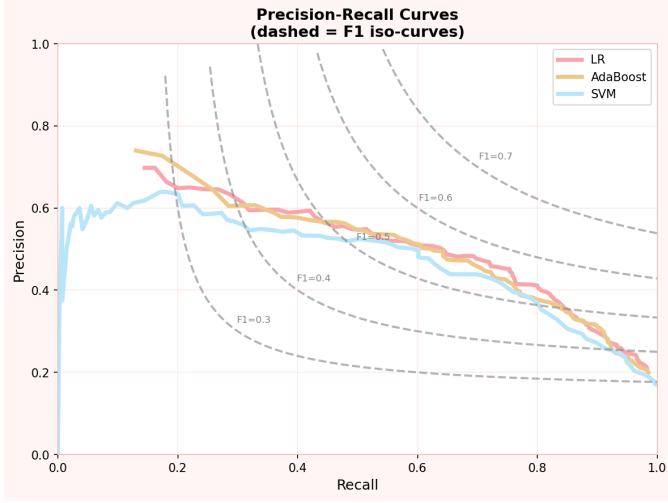


Fig. 8. Precision–Recall curves. Logistic Regression maintains the strongest PR balance.

As shown in Fig. 8, Logistic Regression maintains the strongest precision–recall balance across thresholds, consistent with its superior F_1 -score performance after threshold optimisation. To further examine classification behaviour under the selected thresholds, confusion matrices for all three models are shown in Fig. 9.

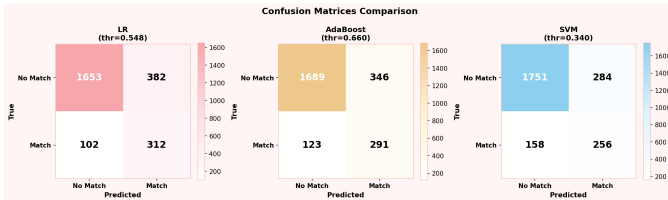


Fig. 9. Confusion matrices under optimal decision thresholds. Logistic Regression correctly identifies ≈ 312 of 414 positive matches.

Importantly, all three models achieve accuracy values of around 81–83%, close to the 83.5% trivial classifier that always

predicts the majority class. This confirms that accuracy is an unreliable metric under severe class imbalance. For example, Logistic Regression identifies approximately 312 out of 414 positive matches while maintaining manageable false positive rates, producing meaningful minority-class prediction.

Most importantly, the relatively small performance gap across models (F_1 range: 0.5367–0.5632) suggests that predictive success is driven more strongly by engineered relationship-level features than by model complexity itself. Despite using only an 18-feature train-only selected subset derived from the shared 54-dimensional engineered feature space, Logistic Regression remains highly competitive with the SVM ensemble, which operates on a 66-dimensional feature representation. Overall, these results indicate that carefully engineered relationship-aware features contribute more substantially to performance gains than increasingly sophisticated model architectures.

C. The Value of Feature Engineering

This section directly evaluates the central hypothesis of the project: engineered relationship-level features are substantially more predictive than raw individual ratings. The correlation-based feature selection stage used in AdaBoost provides a clean ranking of feature importance.

Table III ranks the top features by Pearson $|r|$ with the match label.

TABLE III
TOP-7 FEATURES RANKED BY $|r|$ WITH MATCH LABEL

Rank	Feature	$ r $
1	attractive_pair [engineered]	0.3973
2	funny_pair [engineered]	0.3891
3	like_x_shared_interests_o [engineered]	0.3876
4	trait_pair [engineered]	0.3731
5	like_x_guess_prob_liked [engineered]	0.3247
6	like_iid_centered [engineered]	0.3164
7	like [raw]	0.3047

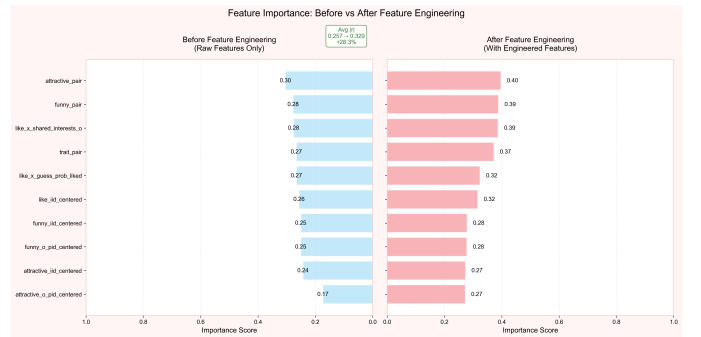


Fig. 10. Top-10 features by $|r|$, colour-coded as Engineered vs. Raw. Engineered features dominate the ranking.

Several important patterns emerge from Fig. 10. The top five predictive features are all engineered rather than raw. `attractive_pair` achieves $|r| = 0.3973$, outperforming

the strongest raw feature (like, $|r| = 0.3047$) by more than 30%. Other engineered features—funny_pair ($|r| = 0.3891$), like_x_shared_interests_o ($|r| = 0.3876$), trait_pair ($|r| = 0.3731$), and like_x_guess_prob_liked ($|r| = 0.3247$)—also rank highly.

These findings demonstrate that successful romantic matching depends more strongly on reciprocal interaction structure than on isolated individual evaluations. A raw attractiveness rating is fundamentally a one-sided signal: it reflects one participant’s perception but ignores reciprocation. In contrast, attractive_pair becomes large only when both participants rate each other highly, directly aligning with the mutual nature of the match outcome. Similarly, like_x_guess_prob_liked operationalises perceived reciprocal liking, which prior behavioural research identifies as one of the strongest predictors of romantic attraction.

Importantly, these interaction features become substantially more reliable after GPB normalisation removes participant-specific scoring bias. By standardising subjective rating scales across individuals, GPB enables pairwise interaction features and cosine-based interest similarity to capture genuine interpersonal compatibility rather than arbitrary differences in scoring behaviour. The dominance of engineered bilateral features therefore supports the proposed relationship-aware feature engineering framework and confirms that interaction-aware representations are central to relationship prediction tasks.

D. Threshold Sensitivity Analysis

Classification thresholds were optimised on validation folds separately for each model. The optimal thresholds differ substantially:

- Logistic Regression: $\tau^* = 0.548$
- AdaBoost: $\tau^* = 0.660$
- SVM ensemble: $\tau^* = 0.340$

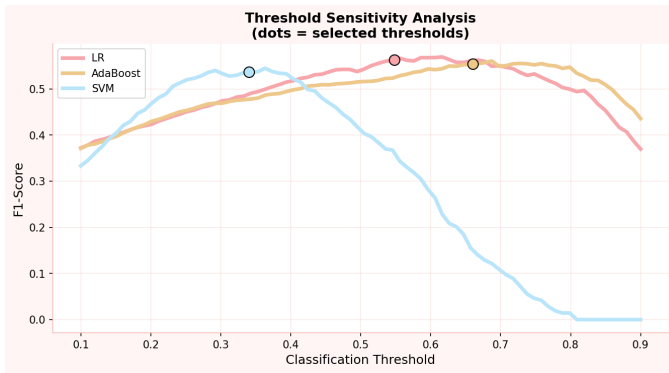


Fig. 11. F_1 -score vs. classification threshold for all three models, with optimal operating points marked.

As shown in Fig. 11, this substantial variation in optimal thresholds does not reflect differences in model quality. Instead, it arises from differences in score calibration and output distribution: Logistic Regression produces relatively

well-calibrated probabilities; AdaBoost generates weighted ensemble scores that require a higher decision threshold; and the Platt-scaled SVM transforms margin-based decision scores into calibrated probability-like outputs. These findings demonstrate that the default threshold of 0.5 should not be treated as universally optimal—under imbalanced classification settings, threshold selection must instead be treated as a tunable hyperparameter specific to each model. Threshold optimisation therefore represents an important component of reliable behavioural prediction under class imbalance.

E. Gender Asymmetries in Feature Importance

Consistent with Fisman et al. [1], the models reveal systematic gender asymmetries in how trait ratings contribute to successful matches. The interaction features gender_attractive and gender_intelligence consistently rank among the more predictive engineered features across the evaluated models. Attractiveness-related interactions appear to contribute more strongly in evaluations made by male participants, whereas intelligence-related interactions exhibit stronger predictive influence in evaluations made by female participants. These asymmetric patterns would remain largely invisible in models relying only on raw participant ratings without gender-aware interaction terms.

More broadly, the consistency between machine learning feature importance patterns and prior behavioural economics findings strengthens the interpretability of the proposed relationship-aware feature engineering framework, suggesting that the learned representations capture meaningful social interaction structures rather than purely statistical artefacts.

F. Computational Efficiency

Training cost differs substantially across models:

- Logistic Regression: 0.35 s
- AdaBoost (110 decision stumps): 3.06 s
- SVM (Top-5 ensemble): 34.07 s
- Full SVM search explored 15 individual candidate models and 64 ensemble combinations, resulting in substantially longer tuning cost

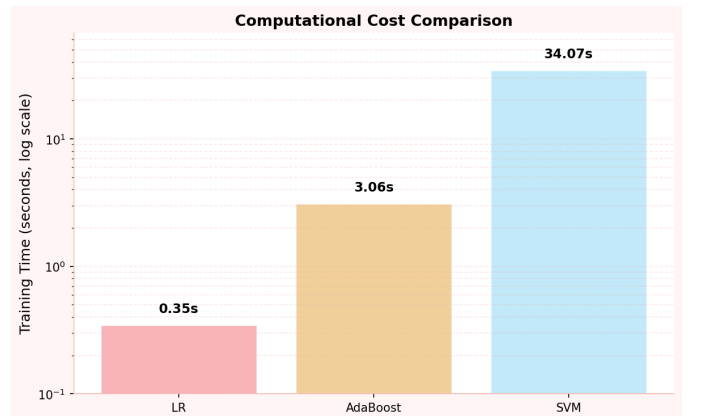


Fig. 12. Training time comparison (log scale).

As shown in Fig. 12, although SVM provides nonlinear modelling capacity, its higher computational cost produces no improvement over Logistic Regression or AdaBoost in the final test comparison. In contrast, Logistic Regression achieves the strongest predictive quality, while AdaBoost remains the most efficient non-linear/ensemble alternative. This trade-off is particularly important in practical deployment settings where frequent retraining may be required as new speed-dating events become available. Overall, Logistic Regression is the strongest final model by F_1 and AUC, while AdaBoost remains a practical alternative due to competitive performance and moderate computational cost.

G. Model Assumptions and Suitability

These three models have different assumptions, which affect their applicability to this task. Logistic regression assumes that the log odds of a match can be approximated by a linear combination of the selected features. Although this is a relatively simple assumption, it works well here because the features we designed already capture key relational patterns, such as reciprocity, bilateral evaluation, and relative preference strength.

AdaBoost is more flexible because it combines many threshold-based weak learners. This allows it to capture nonlinear threshold effects, such as situations where mutual liking only has predictive power after reaching a certain level. However, it may be more sensitive to noisy labels and outliers, especially in cases of class imbalance.

RBF kernel SVM can model more complex nonlinear patterns by measuring similarity in the transformed kernel space. However, it has poor interpretability and higher computational cost. Since it does not outperform simpler models, our results suggest that feature engineering contributes more to performance improvement than increasing model complexity.

V. CONCLUSION, LIMITATIONS, AND FUTURE WORK

A. Conclusion

This study set out to predict mutual romantic match outcomes in speed dating as a supervised binary classification problem. The research mainly formed three interrelated contributions.

Methodological: By rounds, the wave-level, participant-disjoint data split method, effectively prevent the information leakage caused by overlapping the participants. This issue has often been overlooked in previous studies.

Feature engineering: Through correlation analysis, the effectiveness of five types of relationship-level features was verified, including gender interaction, reciprocal, pairwise product, gap, and GPB combined with SIS features. The top five features are all artificially constructed features, among which the correlation of `attractive_pair` reaches $|r| = 0.3973$, significantly higher than any individual raw feature.

Model comparison: Logistic regression on the retention test set has obtained the best overall performance ($F_1 = 0.5632$, AUC=0.8474, 0.35 s). AdaBoost followed closely ($F_1 = 0.5538$, AUC=0.8410, 3.06 s). However, the SVM ensemble, although it needs to handle 66-dimensional input

features and 34.07 s training time, does not perform better than the simpler model ($F_1 = 0.5367$, AUC=0.8229). These results indicate that compared with increasing the complexity of the model, feature engineering contributes more to the predictive performance.

B. Limitations

The dataset was collected in a U.S. graduate school setting from 2002 to 2004, sample population is relatively single, for people with higher education, from western cultures. It is still unclear whether these research models are applicable to other cultures, age groups or contemporary dating environments. In addition, all ratings are self-reported after a four-minute communication under time pressure and may be influenced by social approval bias and the halo effect. This dataset lacks language content, sound features, and non-verbal behavior information, which are likely to carry important compatibility clues. At the same time, the dataset only includes limited interactions in each speed-dating event, making it difficult to fully model broader relationship networks or higher-order interaction patterns. Therefore, the inherent limitations of the dataset itself set an inherent upper limit on the performance that the model can achieve.

C. Future Work

Recommendation and ranking framework. Problem is defined as binary match/no-match, although suitable for academic evaluation, but did not reflect the real world user needs. In practical applications, a participant will interact with multiple potential objects, and what is needed is a preference ranking rather than an isolated binary decision result. A natural extension direction is to redefine the problem as Learning to Rank (LTR). The probability outputs of our model after calibration can be used as continuous ranking signals. Within each event wave, these scores can rank all participants of the opposite-gender for a specific individual, thereby generating a TopK recommendation list that reflects relative compatibility. This way of defining problems directly corresponds to the tasks of commercial dating platforms and is logically based on our existing GPB normalization, which was designed precisely to make cross-person comparisons meaningful.

Graph-based relationship modeling. Previous studies have shown that user interactions in recommendation systems can naturally be represented as graphs [5]–[7]. Similarly, in our task, each participant can be regarded as a node, and each dating interaction can be viewed as an edge between two nodes. In this way, the model can learn not only the features of a pair of participants but also broader relationship patterns across the entire dating network. Within this framework, our existing designed features can still be used. Features such as mutual likes, pairwise rating products, rating differences, and interest similarity can be considered as edge information between participants. Then, the GNN can combine these pairwise features with participant-level information to predict whether a pair of participants is likely to form a mutual match. This extension is more suitable for larger dating datasets, although

future experiments should still avoid participant-level data leakage during the graph construction process.

Temporal and cross-cultural validation. Data sets at the university of Columbia reflects the particular population groups mate preferences. Future research should use data of different natures to verify our main findings, especially regarding the two key findings of gender asymmetry in attractiveness weights and the dominance of reciprocal features. There are two directions that are most important (a) *Temporal generalisation*—using contemporary match data or mobile application date (after 2015), to evaluate in the era of matching algorithm, whether people’s preference patterns have changed; (b) *Cross-cultural generalisation*—using data sets from east Asia or other non-western background, to examine what we find asymmetric preference structure is widespread. If these gender asymmetry patterns cannot be reproduced in different cultural contexts, then the selection of feature engineering needs to be adjusted according to specific cultural backgrounds before global deployment. The above three research directions—ranking-based recommendation, graph-based relationship modeling, and cross-time and cross-cultural validation—provide valuable avenues for extending our current research work. These directions address the main shortcomings of existing binary classification model frameworks, while also pointing the way to expanding the model to larger-scale, more realistic social dating scenarios.

D. Dataset and Code Availability

The dataset used in this project is the Columbia University Speed Dating Experiment dataset, which is publicly available on Kaggle: Kaggle Speed Dating Experiment dataset.

The implementation code is available at: GitHub repository.

The repository contains the source code for data preprocessing, feature engineering, model training, hyperparameter tuning, and model evaluation. The submitted ZIP package also includes the project report, presentation slides, and source code files.

TABLE IV
TEAM MEMBER CONTRIBUTIONS

Name	Role	Contribution
Linchen Wang	Core Contributor	Preprocessing and feature engineering; AdaBoost implementation and tuning; AdaBoost, limitations, and future work writing; feature engineering and logistic regression presentation.
Feixuan Wang	Core Contributor	Logistic regression implementation and tuning; model comparison and visualization; code organization; logistic regression and comparison writing and presentation.
Yingyao Li	Core Contributor	Participated in preprocessing improvement; led SVM optimization and extensive hyperparameter tuning; developed and optimized ensemble models; contributed to SVM section writing, performance analysis, and project progress reporting; presented the SVM and AdaBoost sections.
Yucheng Li	Core Contributor	Preprocessing improvement; literature-based SVM implementation; preprocessing, introduction, performance analysis, limitations, and future work writing; slide organization and preprocessing presentation.
Yunzhen Chen	Supporting Contributor	Preliminary preprocessing; project discussion and presentation preparation; introduction, limitations, and future work presentation.

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